

R Aswin Raj

Generative AI Engineer (LLMs & Agentic Systems)

AGENTIC AI SYSTEMS | ENTERPRISE RAG | LLM ENGINEERING | AGENT ORCHESTRATION | AZURE OPENAI

PROFESSIONAL SUMMARY			
Generative AI Engineer with hands-on experience designing and deploying applied AI solutions , including enterprise-grade LLM, RAG, and agentic AI systems in production environments. Proven expertise in building end-to-end GenAI platforms , including multimodal RAG pipelines, semantic retrieval, prompt frameworks, and agent orchestration using Azure OpenAI and Semantic Kernel . Strong background in applied ML and NLP , with experience operationalizing AI solutions through cloud-native architectures, SQL-backed data pipelines, and robust engineering practices .			
Education	Board	Institute	Year
B.Tech	CSE-AI	Amrita Vishwa Vidyapeetham University	2023
ROLE		WORK EXPERIENCE	Duration
Advanced Application Engineering Analyst		Accenture	Nov 2023 - Present
Roles & Responsibilities	- Designed and deployed enterprise-grade RAG systems integrating SQL data and semantic search to enhance contextual accuracy, improve response consistency, and reduce manual intervention across internal business platforms. - Developed RAG pipelines integrating text and image-based information to support richer insights and improved AI-assisted user experiences. - Designed standardized prompt templates and GenAI workflows enabling consistent, explainable AI responses across multiple enterprise scenarios. - Built agent-based AI systems using Semantic Kernel to automate complex workflows, reduce manual intervention, and support intelligent task orchestration. - Optimized LLM pipelines through chunking strategies, embedding tuning, and retrieval logic to improve inference quality. - Designed SQL-backed data pipelines supporting efficient retrieval and downstream analytics for AI-driven applications. - Enabled production readiness via rate limiting, monitoring hooks, and CI/CD workflows across environments.		
Initiatives & Achievements	- Received 'Star of the Month' award in August 2025 in recognition of outstanding performance and contribution. - Honored with 'Accenture Celebrates Excellence (ACE)' award for exceptional achievements for the FY 2025 & FY 2026.		
INTERNSHIP			Duration
Intern, Computational Linguistics			Pan Lingua May 2022 – Aug 2022
Roles & Responsibilities	- Conducted extensive exploratory data analysis and preprocessing to address imbalanced datasets for training. - Performed statistical estimation techniques ensuring cleaned, balanced data was suitable for model development. - Implemented optimized classification algorithms and built custom models for accurate abusive content detection.		
PUBLICATIONS			
Project 1		Multimodal Assessment of Abusive Content	
Abusive Content Detection using Multimodal Deep Learning model	- Developed classification models detecting abusive comments in Tamil and Tamil-English code-mixed datasets. - Preprocessed multilingual text data and addressed class imbalance using SMOTE to enhance model robustness. - Implemented TF-IDF feature extraction and experimented with SVM using Random Kitchen Sinks effectively. - Evaluated model performance on both datasets and optimized hyperparameters to improve generalization. - Achieved 0.52 F1 score on Tamil-English dataset, showing strong low-resource abuse detection performance.		
Project 2		Indian Company Logo Detection	
Object Detection model using YOLOv3	- Built a computer vision model to detect logos of Indian companies from images with high precision and reliability. - Created a dataset from scratch, ensuring a diverse set of company logos for training and effective validation. - Trained a YOLOv5 model for 100 epochs, optimizing it for real-time logo detection in practical applications. - Developed a Graphical User Interface (GUI) to enable users to upload images and detect logos interactively. - Achieved a mean Average Precision (mAP) of 74.5, ensuring reliable logo recognition across datasets consistently.		
ACADEMIC PROJECT			
Multimodal Assessment of Abusive Content	- Developed a multimodal DL model to detect abusive content in YouTube videos by analyzing both audio and text. - Collected and created a dataset from scratch, ensuring diverse and representative samples for training. - Utilized TensorFlow for deep learning, LibROSA for audio processing, and TF-IDF for text vectorization. - Integrated Transformers for NLP tasks and CNN-LSTM architecture for sequence modeling to improve accuracy. - Achieved a validation accuracy of 0.83, demonstrating the model's effectiveness in detecting abusive content.		
CERTIFICATIONS			
Azure AI Fundamentals (AI-900)	- Learned core AI and ML concepts, gained strong foundation understanding of real-world applications. - Learned implementing the concepts using Azure Machine Learning and Cognitive Services effectively.		2024
Azure AI Engineer Associate (AI-102)	- Gained expertise in designing, building, and managing AI solutions using Azure Cognitive Services. - Acquired proficiency deploying scalable AI systems by Azure Bot Services & support production tools.		2024
SKILLS & INTERESTS			
Skills & Interest	Programming: Python, Azure Services, Async workflows, REST APIs, FastAPI, Vector Databases, Embeddings indexing Machine Learning & AI: Generative AI, Agents, Deep & Reinforcement Learning, NLP, CV, Semantic Search Cloud Technologies: Azure Cognitive Services, Azure Machine Learning, Azure OpenAI, Containerized deployments Tools & Frameworks: Transformers, YOLO, Azure Frameworks, Pandas, OpenCV, Postgre, LLM Frameworks, TensorFlow		